

The Illusion of Optimality: Challenges of increasing complexity in multi-risk assessment

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Abstract

Efforts to integrate multiple hazards, exposure and vulnerabilities into coherent multi-risk frameworks have advanced over the past three decades, yet the field continues to lack operationalisable metrics to assess whether increasing modelling complexity has improved predictive or decision-making efficiency. This paper argues that the bias–variance tradeoff, originally from statistical learning theory, can serve as an analytical lens to interpret this trajectory and to ask whether the field’s growing complexity reflects genuine progress or an “illusion of optimality”. Without explicit measures of informational efficiency, risk integration could become performative rather than functional. We acknowledge that the path toward optimality must be reconciled with the actionability required by practitioners, and that once social components become central to an assessment, what counts as “optimal” is no longer technically determined but becomes relational, context-dependent, and contested. The paper therefore does not prescribe a universal optimum, but questions whether current integration efforts move the field toward or away from informational efficiency—however that efficiency is ultimately defined

1 Introduction

Multi-risk assessments are inherently complex, requiring the integration of compounding hazards, dynamic exposure, and evolving societal vulnerabilities. Over the past three decades, tools and methods have advanced significantly to address these multiple layers of complexity. The bias–variance framework (Geman et al., 1992) provides a pragmatic analytical argument

for a long-standing challenge in risk science: determining when integration enhances understanding and when it merely amplifies uncertainty. Initially developed in machine learning, the concept describes the tradeoff between underfitting (high bias) and overfitting (high variance). Lower predictive error implying a reduced uncertainty over outcomes, then affecting the expected decision (Domingos, 2000). Building on Dunant (2021), who first applied this framework as an analytical argument toward multi-hazard risk assessments, this paper extends the analysis in three directions: (i) a three-phase historical mapping of the field’s trajectory onto the bias–variance continuum, (ii) the introduction of *informational efficiency* as a core evaluative concept, and (iii) a diagnosis of the “illusion of optimality”—the risk that increasing complexity is mistaken for progress in the absence of shared efficiency metrics.

We specifically refer throughout to “multi-risk” frameworks as defined by Zschau (2017): integrated systems that account for complex interactions among multiple hazards, exposed assets, and societal vulnerabilities—as distinct from simpler multi-hazard approaches that consider co-occurrence without systemic interdependencies. It is important to specify that regardless of the terminological scope—whether strictly multi-hazard (physical processes), multi-hazard risk (asset exposure), or multi-risk (systemic societal vulnerability)—the fundamental structural challenge remains analogous with the understanding that the rate of complexity increases at each step. Indeed each step up this hierarchy introduces new parameters, thereby increasing the risk of variance (overfitting) if the underlying process understanding does not scale at the same rate as the model’s complexity.

As Dunant (2021) highlighted, single-hazard models in multi-hazard environments exhibit high bias, while overly complex multi-risk models might show high variance without proportional gains in predictive accuracy (Figure 1). However, this framework has remained largely theoretical. The field lacks operationalisable metrics—quantitative where measurable, qualitative where not—to situate itself within this landscape; whether it sits near an optimal balance or drifts toward inefficiency remains unknown. This paper argues that *informational efficiency*—the capacity to convert increased model complexity into proportional gains in uncertainty-aware, decision-relevant outputs—should become a core evaluative dimension of multi-risk science. The concept draws on decision theory (Horvitz and Barry, 1995), information theory (Shannon, 1948; Cover and Thomas, 2006), and risk communication research (Klinke and Renn, 2002). In the context herein, informational efficiency is distinct from traditional statistical accuracy. In multi-risk environments, a purely quantitative “performance” metric risks missing the local social features and contexts that are fundamental to risk decision-making. Efficiency, therefore, is not a single rigid metric, but encompasses the qualitative utility of the model within its specific social and institutional context. Potential quantification are proposed 1. We acknowledge, that explicitly measuring efficiency may not in itself dissolve the illusion of optimality, particularly when social systems resist the very

notion of a clear-cut and universal optimum. The argument is therefore not that better metrics will solve the problem, but that without any shared language of efficiency, the field cannot even diagnose where it stands

2 Conceptual Framework

The bias–variance framework is used here, as in Dunant (2021), as a conceptual heuristic—not as a literal statistical mapping. In statistical learning theory, bias and variance are mathematically decomposable components of prediction error; in multi-risk science, they are not. The analogy is therefore diagnostic rather than quantitative: it provides a vocabulary for asking whether a given integration effort evolves toward oversimplification (high bias) or toward unmanageable complexity (high variance).

This lens applies beyond physical models. In participative exercises, the pursuit of comprehensive social integration may introduce redundancy, conflicting inputs, or epistemic noise. While the accumulation of heterogeneous and sometimes incommensurable knowledge claims from diverse disciplinary and stakeholder perspectives is a valuable reflection of complex social realities (Schlüter et al., 2024; Rusca et al., 2026), from a purely operational standpoint, it can challenge an actionable synthesis—thereby diminishing informational efficiency without commensurate gains in immediate decision relevance. This complexity also directly undermines interpretability: if practitioners cannot trace the causal logic amidst the noise of competing narratives, they cannot assume the professional responsibility required to translate assessments into action (Pereira Santos et al., 2024).

In this context, the pursuit of “multi-risk” optimality must account for bounded rationality (Simon, 1956, 1947) and acknowledge the absence of “unlimited resources of time, information, and computational might” (Gigerenzer, 2020). Analogous to the “efficiency–resilience” tradeoff (Hynes et al., 2020), hyper-integration (low bias) could produce an “integration–resilience tradeoff”, where noise overtakes the capacity for action.

The following sections attempt to categorise the evolution of multi-risk science into three chronological phases, mapping the field’s trajectory onto the bias–variance continuum (Dunant, 2021), highlighted Figure 1.

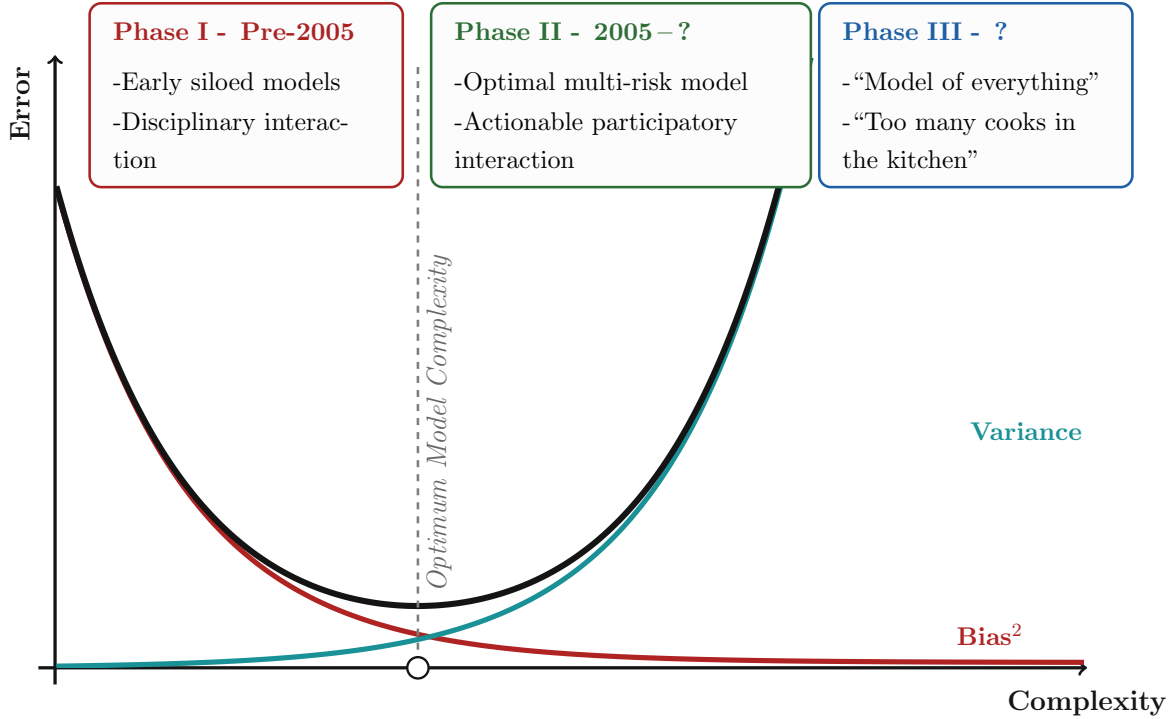


Figure 1: Conceptual framing of the bias-variance landscape for multi-risk assessments.

This framing illustrates the transition from the high-bias simplicity of siloed origins to the high-variance complexity of modern hyper-integration, allowing for a critical evaluation of whether current integration efforts achieve functional efficiency or merely drift toward an “illusion of optimality”.

3 Evolution of multi-risk science

3.1 Phase I: Siloed Origins and Emerging Integration Mandates (before 2005)

The first attempts to formalize multi-hazard assessment emerged alongside early international frameworks promoting integrated disaster management. On the science side, the volcanology community provided significant early precedents in this regard, as volcanic crises inherently involve the interaction of multiple simultaneous hazards like ashfall, lahars, and seismic activity (Tilling, 1989; Blong, 1996).

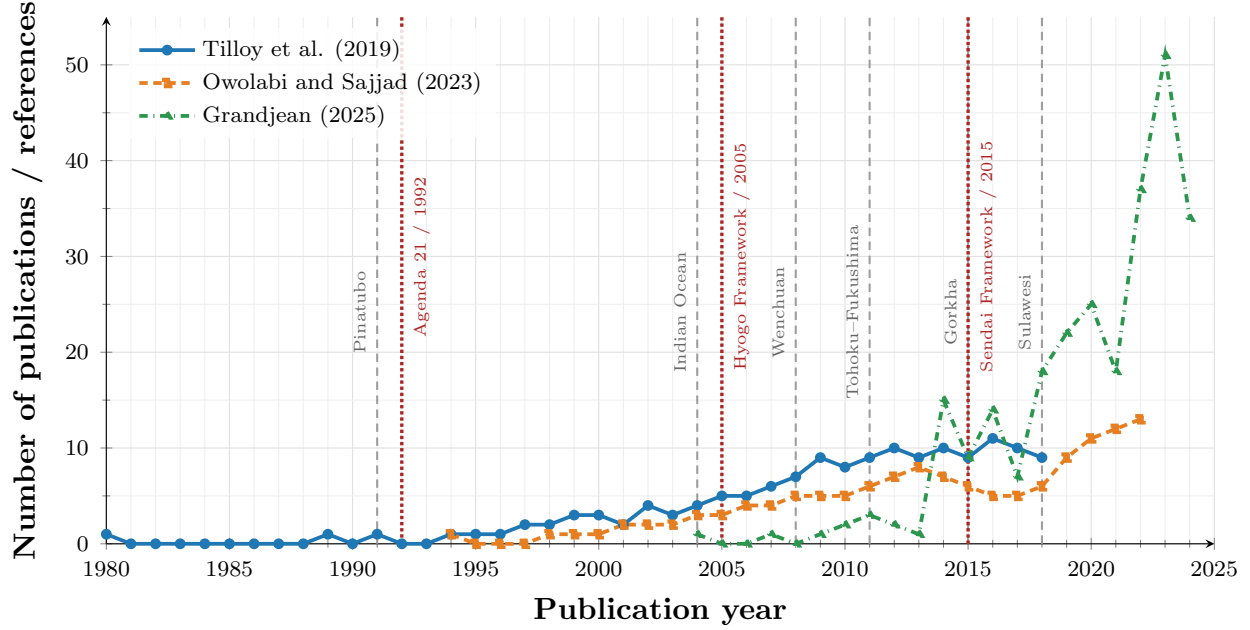


Figure 2: Timeline synthesis of annual publication/reference counts for multi-hazard and multi-risk research. Tilloy et al. counts were rebuilt by aggregating the **Year** column from the supplementary workbook; Owolabi and Sajjad counts were rebuilt by aggregating the **Year** column from the supplementary material; Grandjean is kept as a contextual 2004–2025 annual distribution from Figure 3 in Grandjean (2025), who frames the selected corpus over 2004–2025. Grey dashed lines mark selected major multi-hazard or cascading events since 1990: Pinatubo eruption–lahars (1991), Indian Ocean earthquake–tsunami (2004), Wenchuan earthquake–landslides (2008), Tohoku earthquake–tsunami–Fukushima (2011), Gorkha earthquake–landslides (2015), and Sulawesi earthquake–tsunami–liquefaction (2018). The red dotted line marks the 2005 Hyogo Framework, Agenda 21 in 1992 and the Sendai Framework in 2015.

The principles of sustainable development outlined in *Agenda 21* (UNEP, 1992) and later reinforced by the Johannesburg Plan (Nations, 2002) and the Hyogo Framework for Action (UN-ISDR, 2005) established the groundwork for multi-hazard thinking (Figure 2).

During this phase, risk assessments remained largely disciplinary, emphasizing individual hazards such as floods, earthquakes, or landslides. Studies like Middelmann and Granger (2000) and Granger et al. (1999) in Australia exemplified early local-scale multi-hazard assessments. At this stage most models exhibited high bias: simplified assumptions and a focus on single-event probabilities regardless if they happened in a multi-hazard environment.

The absence of uncertainty quantification and the reliance on heuristic matrices led to low informational efficiency models were interpretable but detached from cascading dynamics

and long-term systemic risks.

3.2 Phase II: Institutional Integration and the Illusion of Optimality (2005–?)

Following the Hyogo Framework, institutional momentum accelerated the integration of multiple hazards. The Sendai Framework (UNDRR, 2015) later reaffirmed this mandate, emphasizing "all-of-society" and "systemic" approaches. These international mandates, by prioritizing 'holistic' and 'all-hazard' frameworks, effectively incentivized the creation of comprehensive systems designed to map every interaction. Conceptual work by Marzocchi et al. (2009) and Kappes et al. (2012) articulated the need for frameworks addressing multi-hazard and multi-risk interactions as single-hazard approaches in multi-hazard environments are inherently biased toward specific features, missing critical interdependencies. The transition from single-hazard to systemic multi-risk assessments is, in part, driven by what Zschau (2017); Hochrainer-Stigler et al. (2024) identifies as the twin challenges of risk globalisation and climate change. Globalisation has rendered risks interdependent through cross-linked critical infrastructure and global supply chains, while climate change increases the frequency and spatial distribution of extreme events (Seneviratne et al., 2021). These drivers create "high-risk dynamics" that require capturing various interactions (Pescaroli and Alexander, 2018), simultaneously stimulating the bias-tradeoff variance dilemma highlighted herein.

Research in the modelling space during this phase reduced bias by linking hazard chains, introducing Bayesian networks, and incorporating network-based or spatio-temporal frameworks (Tilloy et al. 2019; De Angeli et al. 2022; Terzi et al. 2019; Dunant et al. 2021; Dunant et al. 2025). These efforts represented a major step toward a unified understanding of cascading effects and feedback loops. At this stage, multi-hazard adoption induced a rise in variance of the risk models.

Yet the apparent convergence toward optimality might be deceptive. Several reviews show that integration often increased conceptual overlap but still struggle to translate complexity into actionable indicators (Owolabi and Sajjad, 2023; Laino et al., 2024; Hochrainer-Stigler et al., 2024; Scolobig et al., 2017). The field seems to achieve visibility but not necessarily efficiency - defined here as the ability to convert increased model complexity into proportional gains in decision relevance or process understanding. The illusion of optimality persisted not simply because standardized metrics were absent, but because the very notion of what constitutes an informational gain was never made explicit: whether efficiency should be measured in terms of improved risk communication, facilitated action, or deeper process understanding remained unarticulated. This ambiguity—not merely a technical gap—is what could allow increasing complexity to be mistaken for progress. Dunant (2021)'s warning that

as model complexity increases, the risk of "overfitting" escalates, potentially impairing the practical precision of multi-hazard outputs, is even more prevalent today as the tools for large scale computation and collaboration become more accessible (Mack, 2011).

3.3 Phase III: Rising Complexity and the Unquantified Present (?)

While the recent proliferation of highly complex, socially integrated models might suggest the field has already transitioned into a "Phase III" of multi-risk assessment, we argue that this boundary remains entirely unverified. Because the field lacks operationalised metrics for informational efficiency, we simply cannot know whether the current landscape represents a genuine paradigm shift toward optimal risk integration, or merely an inflation of variance and epistemic noise. Consequently, Phase III remains an unquantified horizon rather than a confirmed reality. Nevertheless, the authors believe the field (i.e. multi-hazard, multi-risk) stands at a crossroads today. Advances in computational modelling, remote sensing, and machine learning have enabled "hyper-integrated" frameworks spanning climate, geophysical, and social domains (Ferrario et al. 2025; MYRIAD-EU 2022; Lundstrom and Alam 2022). However, these systems risk overfitting: absorbing massive input data into interconnected systems without tangible measurable improvement in output quality or decision usability.

As demonstrated by recent assessments of European transboundary risk management, the primary hurdle to operationalizing multi-risk science is not a lack of model complexity, but a lack of fundamental harmonization in metrics and data structures across borders (Polese et al., 2024). While scientific "super-models" attempt to simulate complex hazard interactions, practitioners are often still restricted by a "fragmented data landscape" and a lack of common vulnerability classes, which makes the outputs of highly integrated models difficult to compare or legally validate in an operational context (Polese et al., 2024). Furthermore, as Kalogiannidis (2024) highlight, the effectiveness of disaster management in Europe is more significantly hindered by administrative and legislative obstacles—such as the lack of standardized protocols—than by a lack of computational integration. This suggests that the "illusion of optimality" in complex modelling fails to account for the "social reality" of the legal and administrative constraints within which Civil Protection agencies must operate.

Informational efficiency—defined here as the ratio between usable, uncertainty-aware outputs and the complexity of model inputs—remains largely untracked. Without such metrics, integration can drift into epistemic opacity, where uncertainty and redundancy coexist. As Dunant et al. (2021) emphasized, finding a "healthy medium between accuracy (bias) and precision (variance)" requires quantifying the relationship between model complexity, risk

and action.

Future progress depends on being honest about what kind of progress is being made. The objectives of *scientific progress* (advancing process understanding) and *operational progress* (supporting defensible decisions) are not opposed, but they optimise differently and require different standards of evaluation. Conflating them risks producing the “Mushy Middle” (Stonehouse and Snowden, 2007): work that achieves neither genuine process insight nor novel operational application—sophisticated enough to appear rigorous, tractable enough to appear useful, but evaluated against neither standard. For the researcher, the relevant question is therefore not whether a model is complex, but whether its complexity resolves a specific gap in process understanding that simpler models cannot. For the practitioner, the question is whether it narrows the range of defensible decisions under uncertainty in a way that is institutionally actionable (Polese et al., 2024). In both cases, the threshold for “good enough” is as much social and institutional as it is technical—it depends on who will use the output, under what constraints, and toward what decision (Hanger-Kopp et al., 2022; Simon, 1956). What the field currently lacks might not be more integration, but an explicit, shared account of these thresholds. Without it, efficiency claims remain unverifiable, complexity continues to be equated with progress, and the illusion of optimality persists (Scolobig et al., 2017; Kalogiannidis, 2024).

4 Discussion

4.1 Structural Observations

The three phases outlined above trace a field that is moving on a spectrum from high-bias simplicity to high-variance complexity without a clear mechanism for evaluating its actual position and whether this trajectory constitutes progress. Several observations emerge from this mapping.

First, the bias–variance analogy illuminates but does not resolve. In statistical learning, model selection can be guided by cross-validation against held-out data; in multi-risk science, it is challenging to find equivalent ground truth exists against which to calibrate. The analogy therefore functions as a diagnostic vocabulary—useful for framing the problem, but insufficient as a prescriptive tool.

Second, the distinction between scientific and operational progress—raised in Phase III—deserves further attention. The “Mushy Middle” (Stonehouse and Snowden, 2007), originally a concept from competitive strategy describing firms that pursue neither cost leadership nor differentiation, finds a parallel in multi-risk initiatives that achieve neither deep process understanding nor operational tractability. Recognising this risk does not require abandoning

integration, but it does require that researchers and practitioners articulate—before model development—which standard of evaluation applies.

Third, informational efficiency as proposed here remains deliberately underspecified in its measurement. This is intentional: premature formalisation risks privileging one definition of “useful output” over others. Nevertheless, the field would benefit from pilot studies that attempt to quantify efficiency in bounded contexts—for example, by comparing the marginal decision value of adding a second hazard layer to a single-hazard assessment in a specific governance setting. Table 1 summarises candidate metrics across four dimensions for such efforts.

Table 1: Dimensions of informational efficiency for multi-risk assessments. Rather than a single scalar metric, efficiency is operationalised as a multi-dimensional evaluation spanning statistical, decisional, temporal, and institutional domains.

Efficiency Dimension	Core Decision Question	Candidate Metric / Proxy	Theoretical Origin
Information Gain	Does adding this complexity (e.g., a hazard layer) reduce uncertainty about the risk state?	<i>Mutual Information</i> : Measures whether a new model input (e.g., an additional hazard layer) genuinely reduces uncertainty about the risk outcome, or merely adds noise. <i>Variance Reduction</i> : Measures whether the model’s outputs become more stable once that input is included, relative to a simpler baseline.	(Shannon, 1948) (Varley, 2025)
Decision Impact	Does the added complexity actually change the decision that would be made?	<i>Decision Divergence Rate</i> : Frequency with which the complex model triggers a different categorical action (e.g., evacuate vs. shelter) compared to a simpler heuristic. <i>Expected Value of Including Uncertainty (EVIU)</i> : Change in expected loss when acting on the complex model vs. a baseline.	(Horvitz and Barry, 1995) (Klinke and Renn, 2002)
Temporal Efficiency	Does the model yield the decision within the available operational time window?	<i>Decision Latency</i> : Time elapsed from data input to an actionable, interpretable output. <i>Time-to-Decision Window</i> : Ratio of model processing time to the physical lead time of the hazard.	(Reichstein et al., 2025)
Institutional Uptake	Can the end-user actually process and defend the output under pressure?	<i>Utilisation Rate (ITT proxy)</i> : Proportion of offered model runs that are actually used in the final decision (Intention-to-Treat). <i>Cognitive Load</i> : Number of distinct probabilistic inputs a decision-maker must synthesize to reach a binary action.	(Simon, 1956) (Gigerenzer, 2020)

The question then becomes how such pilot studies might be structured in practice—and whether a more systematic evaluative methodology can be designed to test informational efficiency under realistic operational conditions.

4.2 From Theoretical Optimality to Operational Utility

The preceding observations diagnose a structural problem—the conflation of computational complexity with decision-making progress—but they do not prescribe how the field might resolve it. This final section argues that the resolution lies not in building better models, but in rigorously testing whether models, of any complexity, actually improve the decisions they are designed to inform. To make this argument, we import two evaluative frameworks from development economics—the randomised controlled trial (RCT) methodology formalised by Duflo et al. (2006) and the epistemological critique of that methodology by Deaton (2009)—and propose their adaptation to multi-risk science as what we term *Informational Efficiency Trials*.

4.2.1 The Operational Reality of Phase III Models

The core bottleneck exposed by the Phase III landscape is not a deficit of integration but a deficit of uptake. When a civil protection agency is presented with a hyper-integrated, multi-risk model during or in preparation for a crisis, the likely institutional response is not enhanced decision-making but one of two failure modes: paralysis induced by interpretive overload, or quiet reversion to the familiar single-hazard heuristics that the complex model was designed to supersede. Reichstein et al. (2025) frame this structural vulnerability through the early-warning chain’s “five bridges of death”, where the overall efficacy of the system is determined by its weakest link—typically not the hazard forecast itself but the communication and decision-making stages that follow it. This pattern is not unique to disaster risk science. In emergency medicine, Green and Mehr (1997) documented a case at a Michigan hospital where physicians were provided with the Heart Disease Predictive Instrument (HDPI)—a logistic-regression-based chart requiring approximately fifty probability estimates and a pocket calculator—to determine whether patients presenting with chest pain should be admitted to the coronary care unit. Despite the instrument’s statistical sophistication, clinicians largely abandoned it: the cognitive and time costs of interpreting a tool they did not understand exceeded its perceived benefit. A subsequent fast-and-frugal decision tree, using only three sequential binary cues, not only restored physician compliance but outperformed the HDPI in predictive accuracy, reducing both missed heart attacks and false-alarm admissions (Gigerenzer et al., 2011). The parallel to multi-risk assessment is direct. The HDPI’s failure was not one of internal validity—its logistic regression was

well-calibrated—but one of *informational efficiency*: the ratio between usable, actionable output and the cognitive-institutional cost of processing model inputs collapsed to near zero in the operational environment. As Gigerenzer (2020) argues, what determines the quality of a decision under uncertainty is not the complexity of the underlying model but the match between the model’s structure and the decision-maker’s ecological rationality—the cognitive and institutional constraints under which choices are actually made (Endsley, 1995). The lesson for multi-risk science is that a Phase III model, however theoretically comprehensive, achieves zero informational efficiency if its intended users cannot or do not engage with it.

4.2.2 Redefining “Success” via Intention-to-Treat

The RCT framework developed by Duflo et al. (2006) provides a methodological vocabulary for formalising this gap. In clinical and development trials, a fundamental distinction is drawn between two estimands: the *Treatment on the Treated* (ToT), which measures the effect of an intervention on those who actually comply with it, and the *Intention-to-Treat* (ITT), which measures the effect on the entire population to whom the intervention was *offered*, regardless of whether they used it. The ITT is typically lower than the ToT because it incorporates the real-world friction of non-compliance—subjects who were assigned the treatment but did not take it, or did not take it correctly.

Multi-risk models are currently evaluated, implicitly, on a ToT basis: their performance is assessed under the assumption that the end-user perfectly understands, correctly implements, and fully acts upon the model’s outputs. This is the equivalent of evaluating a drug’s efficacy only among patients who took every dose on schedule. An ITT-based evaluation would ask a fundamentally different question: what is the actual impact on disaster preparedness, response time, or loss reduction when a complex multi-risk model is *offered* to a civil protection agency—factoring in the realistic probability that the agency misinterprets the probabilistic outputs, lacks the computational infrastructure to run the model in real time, or simply defaults to pre-existing protocols under time pressure? If the ITT effect is zero—if the model, once offered, produces no measurable improvement in operational outcomes—then its informational efficiency is zero, regardless of its mathematical elegance or its performance under controlled conditions.

This reframing has a concrete implication for the bias–variance tradeoff. A model sitting at the theoretical optimum of the bias–variance curve (low total error under perfect conditions) may in practice be outperformed—on an ITT basis—by a simpler, higher-bias model whose outputs are immediately interpretable and institutionally actionable. The “illusion of optimality” is therefore not merely a theoretical concern about unmeasured variance; it is a practical concern about unmeasured non-compliance. The field’s bias–variance landscape, as described in Section 2, must be extended to include a third axis: the probability of effective

institutional uptake.

4.2.3 Uncovering the Mechanism of Decision-Making

However, adopting an ITT framework without further qualification would itself be insufficient. Deaton (2009) provides a critical counterweight to the RCT paradigm by arguing that average treatment effects—whether ITT or ToT—are epistemically shallow if they are not accompanied by an understanding of *mechanisms*. In Deaton’s framing, knowing *that* an intervention works (or fails) is of limited scientific value if we do not know *why* it works or fails and *for whom*. The risk, as Deaton warns, is that the methodological apparatus of randomisation becomes a substitute for theoretical understanding—a form of “technical credibility” that displaces substantive inquiry into causal pathways.

For multi-risk science, this caution translates into a specific research imperative: it is not enough to conduct behavioural RCTs that compare decision outcomes when agencies are given complex versus simple models. We must simultaneously map the institutional and cognitive pathways through which model outputs are (or are not) translated into action. Where, precisely, does the communication breakdown occur? Is it a deficit of statistical literacy among local decision-makers who must interpret probabilistic outputs as categorical actions? Is it a structural mismatch between the continuous probability distributions produced by multi-risk models and the binary legal architectures (“declare state of emergency” / “do not declare”) within which civil protection operates (Polese et al., 2024)? Is it a temporal mismatch—the model requires hours of post-processing while the decision window is measured in minutes (Reichstein et al., 2025)? Or is it, as bounded rationality would predict (Simon, 1956; Gigerenzer, 2020), that the sheer number of interacting variables in a Phase III model exceeds the cognitive capacity of any individual decision-maker, forcing a reversion to satisficing heuristics regardless of the information made available?

Without answers to these mechanism-level questions, even a well-designed ITT evaluation risks producing what Deaton terms “uninformative” results: we might learn that a complex model “does not work” in a given agency, but not whether the failure is remediable (through training, interface redesign, or institutional reform) or structural (inherent to the mismatch between model complexity and decision-making architecture). The Dufflo–Deaton synthesis thus points toward a specific methodological programme: randomised evaluations of model uptake that are designed *ex ante* to identify the causal mechanisms of failure, not merely its average magnitude.

4.2.4 Informational Efficiency Trials

The preceding arguments converge on a concrete proposal. We propose that the next frontier of multi-risk science is not the construction of a “Phase IV” model that adds yet more variables, layers, or coupling mechanisms but by the design of *Informational Efficiency Trials* (IETs). IET would be controlled, collaborative experiments in which model developers and end-users rigorously test how varying levels of model complexity affect the speed, accuracy, and defensibility of real-world emergency decisions.

An example of an IET would proceed as follow. A set of civil protection agencies, matched on institutional capacity and hazard exposure, would be randomly assigned to receive either (a) a complex multi-risk model output, (b) a simplified derivative of that output (e.g., a fast-and-frugal decision tree distilled from the full model, following Gigerenzer et al. 2011), or (c) no model-based input (the status quo baseline). Agencies would then participate in standardised crisis simulation exercises, and the primary outcomes—decision latency, accuracy relative to a known ground-truth scenario, and the institutional defensibility of the chosen action—would be measured and compared across arms. The ITT estimate would capture the real-world effect of *offering* each level of complexity; the ToT estimate would capture the effect conditional on actual use; and qualitative process tracing, embedded within the trial design following Deaton’s methodological prescription, would identify the specific institutional bottlenecks that mediate the gap between the two.

This agenda does not dismiss the value of complex, deeply integrated models. It does, however, insist that their value be demonstrated empirically rather than assumed theoretically. And it introduces a possibility that the field might have been reluctant to entertain: that the optimal point on the bias–variance curve, once institutional uptake is factored in, may lie closer to Phase II than Phase III. Not because simpler models are “better” in any absolute sense, but because the marginal gains in theoretical accuracy from hyper-integration may be more than offset by the marginal losses in institutional actionability. If this turns out to be the case, the field’s primary task would shift from reducing model bias to reducing the *implementation variance* that prevents even well-calibrated models from reaching the decisions they are meant to inform.

5 Conclusion

The history of multi-risk assessment reflects oscillations between bias and variance rather than a smooth progression towards optimality. This perspective suggests that increasing model complexity does not necessarily improve understanding or decision relevance, as integration often introduces additional uncertainty and interpretability challenges. Recent advances in holistic frameworks and AI-driven systems highlight both the potential and the

difficulty of managing these trade-offs (Ward et al., 2026; Reichstein et al., 2025).

Building on the bias–variance perspective introduced by Dunant (2021), this paper argues that the central challenge is not only methodological but epistemic. Greater integration must be accompanied by clearer research intent and a systematic evaluation of informational efficiency—namely, how uncertainty, interpretability, and usability evolve with complexity. The bias–variance framework therefore serves not as an end, but as a diagnostic tool to assess whether added complexity yields meaningful gains.

To address this, we propose Informational Efficiency Trials (IETs) as a structured approach to compare modelling strategies under controlled conditions. By explicitly testing how different levels of complexity affect uncertainty propagation, interpretability, and decision usefulness, IETs provide a practical mechanism to determine when integration enhances—and when it undermines—informational efficiency.

Establishing such approaches is essential to ensure that the evolution of multi-risk modelling remains both rigorous and actionable in the context of increasingly interconnected global risks.

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Competing Interests

The authors declare no competing interests.

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